The Impact of Discretization on Estimation and Inference of Structural Models

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Background

Dynamic programming (DP) is widely used in economics to study sequential decision-making under uncertainty, and substantial work has been devoted to developing numerical methods to solve such models. The goal of this research project is to further our understanding of how the computational methods based on *discretization* influence econometric analysis in DP settings. Despite the widespread use of discretized state spaces, comparatively little attention has been devoted to the downstream econometric consequences of numerical approximation. Meanwhile, the consequences may be dramatic for some of the parameter estimates. In a seminal paper, Rust (1987) for example notes (p.1024) that while discretization may have little effect on the approximated value function, the parameter estimates of the structural model may change significantly depending on the choice of grid size.

The literature on discretization is vast and spans a variety of fields ranging from applied mathematics, engineering and computer science. In economics, an important contribution is that of Tauchen (1986) who proposes finite state Markov chain approximations to finite lag VAR models. The key idea is to use an equally spaced grid that takes into account the mean and standard deviation of the marginal distribution of the process, as the support of the finite state Markov chain. The transition probabilities of the chain are then fitted so as to match those of the continuous process. This method remains widely applied in dynamic economic models, including studies of capital misallocation (Asker, Collard-Wexler, and Loecker, 2014), investment heterogeneity (Bachmann and Bayer, 2014), labor market volatility (Amaral and Tasci, 2016), government debt (Adam and Grill, 2017), firm dynamics (Ayres and Raveendranathan, 2023), and consumption smoothing puzzles (Adams and Rojas, 2024). This approach has been extended to general nonlinear, non-Gaussian Markov processes.

As already pointed out, there are relatively few works that address the impact of discretization on the econometric analysis of structural models. A key reference is Fernández-Villaverde, Rubio-Ramírez, and Santos (2006), who studied likelihood-based inference when the model solution is available only numerically. They established that approximated likelihood converges pointwise to exact likelihood as grids are refined (for fixed sample length) under technical regularity conditions (continuity/differentiability, uniqueness and absolute continuity of the invariant distribution, well-defined mapping from shocks/states to observables, etc.). Without the conditions, likelihood convergence can fail even if policy functions converge. Therefore, their practical recommendation is to refine the numerical solution until improving accuracy no longer meaningfully changes the approximate likelihood. This result supports the use of computed likelihoods in principle, but it does not by itself settle how discretization affects parameter inference in practice.

For the remaining results in Fernández-Villaverde, Rubio-Ramírez, and Santos (2006), Ackerberg, Geweke, and Hahn (2009) provide a follow-up comment demonstrating, via a counterexample, that the likelihood-based error bound derived in Fernández-Villaverde, Rubio-Ramírez, and Santos (2006) is incorrect. They argue that inference should be assessed based on the behavior of the estimator itself (i.e., the argmax/argmin), rather than on discrepancies in the level of the joint

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likelihood function. Their analysis shows that approximation errors do not have first-order effects on parameter inference, nor do they accumulate over time. Instead, the resulting bias in parameter estimates is bounded by a term of the same order as the policy function approximation error and, importantly, does not depend on the sample size T. In contrast, differences in joint likelihood values do not share this property and may exhibit different scaling behavior. As a consequence, consistency of the estimator requires only that the approximation error vanishes asymptotically, regardless of the rate. Moreover, to ensure asymptotic normality, it is sufficient that the approximation error converges to zero at a rate faster than $\frac{1}{\sqrt{T}}$, rather than $\frac{1}{T}$.

These results, however, are asymptotic in nature. As emphasized by the authors, "one's estimates are only going to be as accurate as one's approximations, and we believe one should make concerted efforts to make these approximations as precise as possible." While this literature provides a theoretical foundation highlighting the importance of discretization, it offers limited guidance on the practical application. In particular, it does not establish which grid specifications are sufficient to ensure economically negligible estimation error, nor does it quantify the sensitivity of preference parameter estimates and moment conditions to refinements of the state space.

Parallel to this literature is work that studies approximation of the Bellman operator directly; a thorough survey is provided by Rust (1996).

Motivation

While researchers understand that finer grids improve solution accuracy, the practical downstream consequences for parameter estimation remain poorly understood. Standard grid choices in applied work may introduce economically meaningful bias, distorting moment conditions used in GMM estimation.

Research Question

This project investigates how discretization choices in solving dynamic programming problems affect parameter estimates in structural models. Specifically: Does the density of state space grids introduce systematic bias in GMM estimation of preference parameters? How does this bias scale with grid refinement? What grid specifications are sufficient to achieve economically negligible estimation error?

Contributions

This research makes three contributions: 1) First systematic documentation of discretization-induced GMM bias in life-cycle models, bridging numerical analysis and simulation-based estimation literature; 2) Practical guidance for applied researchers on the refinement of the grid; 3) Methodological insight that numerical approximations interact with estimation procedures in subtle ways, deserving additional scrutiny.

Expected Outputs

- Bias-variance decomposition across grid specifications and sample sizes
- Convergence rate estimates for bias reduction under grid refinement
- Recommended minimum grid densities for standard life-cycle models

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